People I Know: Job Search and Social Networks

Cingano, F., & Rosolia, A. (2012). Journal of Labor Economics.
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Overview

- Information is essential during job search
  - Market conditions, job opportunities, etc.
- Does insider information facilitates job search? By how much?
  - Knowing someone nears the job opportunity → reduces search costs → shorter unemployment spell

This paper:

- Estimate the causal effect of **network employment rate** on **unemployment duration**
- Mechanism: **information sharing**

We focus on identification strategy and how authors justify their causal interpretation

Empirical model

Consider a model of network effects on unemployment duration:

$$u_i = \alpha + \gamma ER_{it_0} + \theta \log(N_{it_0}) + X_{it_0}\beta + e_{it_0}$$

- u_i: (log) unemployment duration of invidivual i
- ER_{it_0}: network employment rate at the time of displacement t_0

Baseline definition of network W_{it_0}:

- Coworkers of individual i in $[t_0 - 5, t_0]$ with at least 1 month of overlap
  - Past coworkers network provide more valuable job information than other networks
- Individual-specific: key to the identification strategy

Identification challenges

$$u_i = \alpha + \gamma ER_{it_0} + \theta \log(N_{it_0}) + X_{it_0}\beta + e_{it_0}$$

Problem: ER_{it_0} is non-random

The relationship between ER_{it_0} and u_i has various sources of endogeneity:

1. Self-select to quit jobs
2. Unobserved worker characteristics
3. Common local labor market shocks

1. Self-select to quit jobs
    - Workers with better ER_{it_0} are more likely to quit and have lower u_i
    - Biased sample: only observe workers who *actually* quit
    - **Solution**: exogenous job displacement – firm shutdown
        - Workers in the same shutdown event are dropped from W_{it_0}
2. Unobserved worker characteristics
    - High-ability workers have better ER_{it_0} and lower u_i
    - **Solution**: (i) closing firm fixed effects; (ii) control for observable employment history
        - Make comparison across workers within the same shutdown firm
            - Absorb unobserved dimensions that appear in sorting behavior
        - Control wage history and tenure, which proxies for ability

3. Common local labor market shocks
    - Worker and her coworkers are subject to common labor market shocks
    - Think about workers’ network in Hsinchu Science Park when TSMC is booming
    - **Solution**: add labor market fixed effects

Wrap up

1. Exogenous job displacement: Firm shutdown
2. Unobserved worker characteristics:
    1. Closing firm fixed effects: within-closing-firm comparison
    2. Observable individual and network characteristics: wage history, tenure, etc.
    3. **Caveat**: unobservables not revealed through sorting are not controlled (e.g. differential skill accumulation)
3. Local labor demand shocks: local labor market (LLM) fixed effects

Data

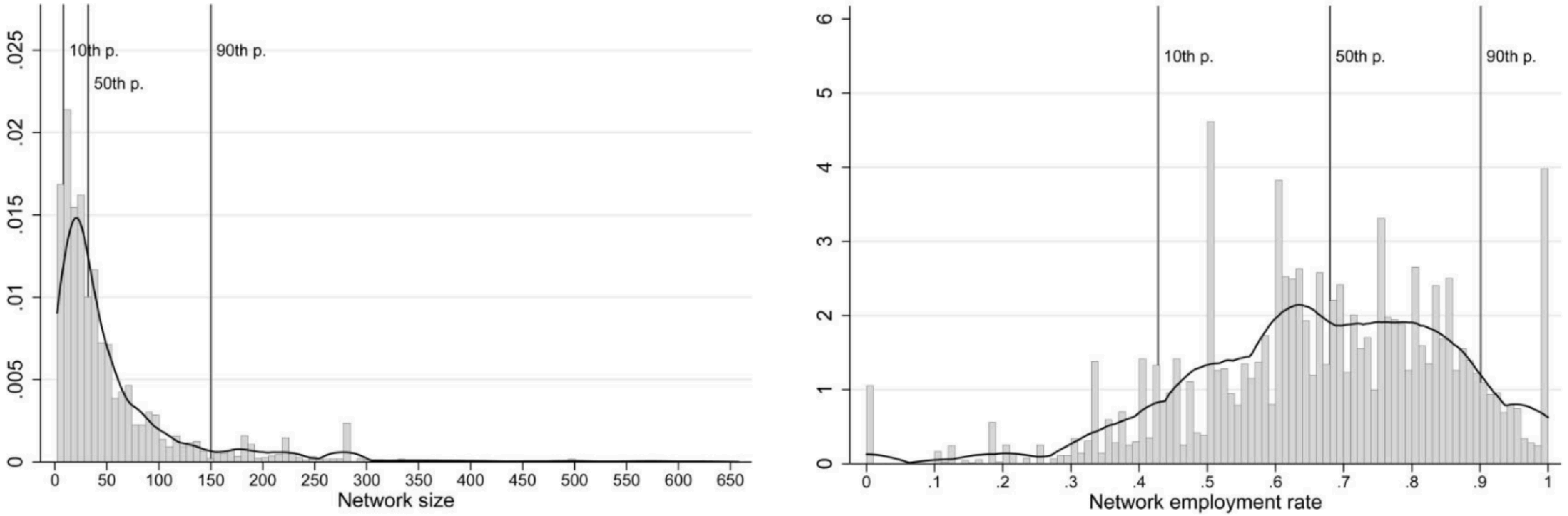
- Employer-employee linked panel from 2 Italian provinces, 1975-1997
  - Demographic: age, gender, place of birth, residence
  - Job & employer information: weekly earnings, tenure, industry, location
  - Taiwanese equivalent: labor insurance data
- Consider only shutdown events in 1980-94
  - 5-year window for network definition
  - 3-year window for unemployment spell completion
    - median = 5 months; mean = 10 months; minimal censoring

Table 1
Closing Firms and Codisplaced Workers: Descriptive Statistics

	Percentile 10th (1)	Percentile 50th (2)	Percentile 90th (3)	Mean	Standard Deviation
Number of codisplaced workers	1	5	15	7.6	10.2
Average age	20	27	38	28	7
% male	0	66.7	100	57.1	39.8
% blue collar	0	100	100	82.0	32.8
% live in:					
Same LLM as closing firm	14.3	88.9	100	76.0	31.8
Same town as closing firm	0	33.3	100	38.2	33.2

NOTE.—Table entries are the relevant statistics computed on the sample distribution of the closing firm–level row variable. Codisplaced workers are defined as those working in the closing firm in the last month of activity.

Summary statistics (Cingano and Rosolia 2012, Table 1)



Network size distribution

Network employment rate distribution

Figure 1: Network characteristics (Cingano and Rosolia 2012, Figure 1)

Baseline results

Table 2
Unemployment Duration

	(1)	(2)	(3)	(4)	(5)
(Log) network size	−.027	.022	−.029	−.057	−.066
	(.017)	(.020)	(.038)	(.044)	(.047)
Network employment rate	−.294*	−.314**	−.385**	−.336*	−.348*
	(.120)	(.120)	(.126)	(.146)	(.125)
Wage at displacement		−.231**	−.228**	−.235**	−.290**
		(.060)	(.059)	(.066)	(.073)
Predisplacement wage growth		.133	.130	.0673	.197
		(.108)	(.110)	(.120)	(.132)
Predisplacement unemployment		.398**	.521**	.446**	.433**
		(.083)	(.105)	(.119)	(.129)
No. employers prior to displacement					
−1			−.274**	−.347**	−.390**
			(.092)	(.105)	(.115)
−2			−.188**	−.244**	−.287**
			(.066)	(.078)	(.086)
−3			−.073	−.110	−.145^{+}
			(.062)	(.074)	(.082)
Average size of firms prior to displacement			.039	.064	.069
			(.049)	(.056)	(.062)
Average commuted distance prior to displacement			−.005	.060	−.031
			(.037)	(.121)	(.133)
Closing firm fixed effects	Y	Y	Y	Y	Y
Year × LLM	N	N	N	Y	N
Year × three-digit sector experience	N	N	N	Y	N
Year × two-digit sector experience × LLM	N	N	N	N	Y
Town and three-digit sector fixed effects	N	N	N	N	Y
Observations	9,121	9,121	9,121	9,121	9,121

Baseline results (Cingano and Rosolia 2012, Table 2)

LHS: log unemployment duration

Estimated elasticity: -0.3

10 ppt ↑ network employment rate

⇒ 3% ↓ unemployment duration

⇒ (1.2 weeks ↓)

Interpretation

Network effects ≠ Information sharing effects

1. **Information sharing**: Someone proximated to job opportunities shares those information
    - search costs ↓ ⇒ duration of job search ↓
2. **Peer pressure**: Knowing most of your friends are working, you feel more pressure to find a job
    - reservation wage ↓ ⇒ duration of job search ↓
3. **Expectation**: Knowing most of your friends are working, you think the labor market is booming
    - reservation wage ↑ ⇒ duration of job search ↑

Authors conduct falsification tests to reject (2) and (3)

Falsification excercises

2. **Peer pressure**:
    1. Use lag network employment rate: if pressure exists, should be a significant predictor
    2. Control for contacts' ability: if pressure exists, $\hat{\gamma}$ should become smaller
        - Proxies for contacts' ability: (i) wage (ii) tenure
3. **Expectation**:
    - Model the expectation on contacts' employment using a probit model: $\Pr(E_{jt_0}) = \Phi(\text{age, time, location})$
    - Get $\widehat{\Pr}E_{jt_0}$, take average over contacts, and use it as a control variable
    - If expectation matters, $\hat{\gamma}$ should become smaller

Table 3
Alternative Mechanisms

		Past Employment		Contacts' Ability			Expected Employment		
	Baseline	−2 Years	−3 Years	Wage	Wage Fixed Effects	Employer Fixed Effects			Entry Wage
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(Log) network size	−.057	−.068	−.066	−.050	−.059	−.058	−.068	−.054	.013
	(.044)	(.044)	(.044)	(.044)	(.044)	(.044)	(.044)	(.044)	(.014)
Network employment rate	−.336*			−.382**	−.319*	−.356*		−.365*	−.051
	(.146)			(.148)	(.150)	(.174)		(.165)	(.049)
Past employment rate		.112	.011						
		(.170)	(.170)						
Contacts' ability				.278+	−.112	.042			
				(.150)	(.218)	(.188)			
Expected employment rate							−.253	.151	
							(.340)	(.381)	
Wage at displacement	−.235**	−.236**	−.235**	−.242**	−.234**	−.235**	−.237**	−.237**	.144**
	(.066)	(.066)	(.066)	(.066)	(.066)	(.066)	(.066)	(.066)	(.024)
Predisplacement unemployment	.446**	.411**	.402**	.455**	.445**	.447**	.424**	.439**	−.099*
	(.119)	(.119)	(.119)	(.119)	(.119)	(.119)	(.119)	(.122)	(.041)
Observations	9,121	9,121	9,121	9,121	9,121	9,121	9,119	9,119	9,121

Alternative mechanisms (Cingano and Rosolia 2012, Table 3)

Information availability

Some contacts are more likely to hold valuable job informations:

- Switch jobs recently
- Work in the same industry / nearby workspace
- Live nearby
- Stronger ties (being coworkers for a longer time)

Table 4
Information Availability and Circulation

	Baseline (1)	Contacts' Turnover (2)	Distance from Contacts': 2-Digit (3)	3-Digit (4)	Workplace (5)	Residence (6)	Ties' Intensity (7)	Propensity to Share (8)
(Log) network size	−.057	−.082^{+}	−.058	−.057	−.057	−.057	−.060	−.057
	(.044)	(.045)	(.044)	(.044)	(.044)	(.044)	(.044)	(.044)
Network employment rate	−.336*							−.336*
	(.146)							(.146)
Movers (M_i/N_i)		−.581**						
		(.170)						
Stayers (S_i/N_i)		−.252^{+}						
		(.150)						
Close (CE_i/N_i)			−.426**	−.373*	−.371*	−.328*		
			(.150)	(.150)	(.150)	(.150)		
Far (FE_i/N_i)			−.124	−.298^{+}	−.267^{+}	−.345*		
			(.170)	(.160)	(.150)	(.150)		
Strong (SE_i/N_i)							−.445**	
							(.170)	
Weak (WE_i/N_i)							−.307*	
							(.150)	
Match quality								−.061
								(.194)
Wage at displacement	−.235**	−.232**	−.231**	−.234**	−.236**	−.235**	−.231**	−.235**
	(.066)	(.066)	(.066)	(.066)	(.066)	(.066)	(.066)	(.066)
Predisplacement unemployment	.446**	.436**	.436**	.441**	.444**	.447**	.372**	.447**
	(.119)	(.120)	(.120)	(.120)	(.120)	(.120)	(.130)	(.119)
Observations	9,121	9,121	9,121	9,121	9,121	9,121	9,121	9,121

Information availability (Cingano and Rosolia 2012, Table 4)

Heterogeneity by network composition

Table 5
Network Composition: Qualification, Age, and Sex

	(1)	(2)	(3)
Network size	.020	.023	.019
	(.020)	(.020)	(.020)
Past unemployment	.406**	.408**	.396**
	(.083)	(.084)	(.083)
Wage at displacement	−.230**	−.232**	−.228**
	(.060)	(.060)	(.060)
Employment rate:			
Same qualification	−.336**		
	(.125)		
Different qualification	−.110		
	(.188)		
Same cohort		−.329*	
		(.135)	
Different cohort		−.276*	
		(.140)	
Same sex			−.024
			(.142)
Different sex			−.379**
			(.125)
Observations	9,121	9,121	9,121

Network composition (Cingano and Rosolia 2012, Table 5)

Indirect connections

- **Competitor**: Contacts' contacts may compete with you for the same job
- **Provider**: Contacts' contacts may help you find a job
- **Diverse contacts** provide less duplicated information

Table 6
Indirect Connections

	Baseline (1)	Indirect Networks: Competitors (2)	Indirect Networks: Providers (3)	Networked Firms (4)
(Log) network size	−.057	−.014	−.058	.044
	(.044)	(.047)	(.044)	(.064)
Network employment rate	−.336*	−.276+	−.337*	−.334*
	(.146)	(.150)	(.150)	(.150)
No. competitors		.007*		
		(.003)		
No. indirect links			.003	
			(.027)	
Networked firms				−.146*
				(.067)
Wage at displacement	−.235**	−.235**	−.235**	−.233**
	(.066)	(.066)	(.066)	(.066)
Predisplacement unemployment	.446**	.475**	.447**	.421**
	(.119)	(.120)	(.120)	(.120)
Observations	9,121	9,121	9,121	9,121

Indirect connections (Cingano and Rosolia 2012, Table 6)

Takeaways

- **Information sharing** in social networks facilitates job search
  - **Positive spillover** of one’s employment status to network members
- **Identification strategy**:
  - Exogenous job displacement
  - Closing firm fixed effects
  - Observable individual and network characteristics
  - Local labor market fixed effects